

A Closed-Subalgebra Counterexample to the Composition-Automorphism Conjecture

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Source conjecture

Behera–Maurya–Muthukumar [1] define an algebra automorphism to be a bijective linear multiplicative self-map. They explicitly allow subalgebras of $H^\infty(\mathbb{D})$ which are not closed. Their concluding conjecture is:

Every algebra automorphism of every subalgebra of $H^\infty(\mathbb{D})$ is a composition operator induced by a disc automorphism.

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Concluding Remarks: In Section 4, we proved that every automorphism of ψH^∞ or $\psi A(\mathbb{D})$ is a composition operator induced by a disc automorphism. We also proved in Section 2 that every automorphism of $\mathcal{B}_a^n$ or $\tilde{\mathcal{B}}_a^n$, which are not of the form ψH^∞ , is also a composition operator induced by a disc automorphism. It leads us to expect that every automorphism of any subalgebra of H^∞ can be extended to an automorphism of H^∞ , and hence it will be composition operator. In view of this, we make the following conjecture.

Conjecture: Let $\mathcal{A}$ be any subalgebra of bounded analytic functions H^∞ on the unit disc in the complex plane. Every algebra automorphism of $\mathcal{A}$ must be a composition operator induced by some disc automorphism.

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REFERENCES

[1] K. BEHERA, R. MAURYA and P. MUTHUKUMAR, *Composition operators and automorphisms of subalgebras of H^∞* , *Journal of Mathematical Analysis and Applications*, 534 (2024), 109701.

The construction below disproves the conjecture even after imposing closedness and requiring the automorphism to be unital and isometric.

Idea of the proof

The map $z \mapsto z^m$ realizes the disc as an m -sheeted quotient of itself. Pullback along this map identifies $H^\infty(\mathbb{D})$ isometrically with the closed subalgebra of functions depending only on z^m . Every disc automorphism acts on the quotient algebra, so conjugation gives an automorphism of this subalgebra. A non-rotational Möbius map does not lift through the branched cover $z \mapsto z^m$: its pullback has m simple zeros, whereas an m -th power can only have zeros whose multiplicities are divisible by m .

For $a \in \mathbb{D}$, write

$$\tau_a(w) = \frac{a - w}{1 - \bar{a}w}.$$

Theorem 1 (power-pullback counterexamples). *Fix an integer $m \geq 2$ and $a \in \mathbb{D} \setminus \{0\}$. Define*

$$\mathcal{A}_m = \{g(z^m) : g \in H^\infty(\mathbb{D})\}.$$

Then $\mathcal{A}_m$ is a closed unital subalgebra of $H^\infty(\mathbb{D})$, and

$$T_{a,m}(g(z^m)) = g(\tau_a(z^m)) \quad (1)$$

is a unital isometric algebra automorphism of $\mathcal{A}_m$. There is no analytic self-map $\varphi : \mathbb{D} \rightarrow \mathbb{D}$ such that

$$T_{a,m}f = f \circ \varphi \quad (f \in \mathcal{A}_m).$$

In particular, $T_{a,m}$ is not induced by a disc automorphism.

Proof. Let $p_m(z) = z^m$ and define

$$J_m : H^\infty(\mathbb{D}) \longrightarrow H^\infty(\mathbb{D}), \quad J_m g = g \circ p_m.$$

Because $p_m(\mathbb{D}) = \mathbb{D}$,

$$\|J_m g\|_\infty = \sup_{z \in \mathbb{D}} |g(z^m)| = \sup_{w \in \mathbb{D}} |g(w)| = \|g\|_\infty.$$

Thus J_m is an isometric unital algebra monomorphism. Its range is exactly $\mathcal{A}_m$, and an isometric range is closed.

Composition by τ_a is an isometric algebra automorphism of $H^\infty(\mathbb{D})$. Therefore

$$T_{a,m} = J_m C_{\tau_a} J_m^{-1}$$

is a well-defined unital isometric algebra automorphism of $\mathcal{A}_m$, and the displayed formula is precisely (1).

Suppose, toward a contradiction, that $T_{a,m} = C_\varphi|_{\mathcal{A}_m}$ for some analytic self-map φ of $\mathbb{D}$. Apply both maps to the element $u(z) = z^m = J_m(w)$ of $\mathcal{A}_m$. We obtain

$$\varphi(z)^m = \tau_a(z^m). \quad (2)$$

The zeros of the right side are exactly the m solutions of $z^m = a$. They are distinct and nonzero. Since τ_a has a simple zero at a and $mz^{m-1} \neq 0$ at each of these roots, every zero of $\tau_a(z^m)$ is simple.

On the other hand, if φ has a zero of multiplicity $r \geq 1$, then φ^m has a zero of multiplicity mr . Hence every zero multiplicity on the left side of (2) is divisible by $m \geq 2$. This contradicts the simple zeros on the right side. No such φ exists. $\square$

Remark 2 (the elementary nonclosed version). *The literal conjecture already fails on the polynomial subalgebra $\mathbb{C}[z] \subset H^\infty(\mathbb{D})$: the map $p(z) \mapsto p(z/2)$ is an algebra automorphism with inverse $p(z) \mapsto p(2z)$, but $z/2$ is not a disc automorphism. Theorem 1 shows that nonclosedness or discontinuity is not the issue.*

Scope, verification, and novelty status

The theorem is a direct counterexample to the source conjecture as stated. It also rules out the natural repairs “closed unital subalgebra” and “continuous/isometric automorphism.” It does not classify which embedded copies or quotient-type subalgebras have the extension property.

The proof is purely algebraic and uses only surjectivity of $z \mapsto z^m$, the standard automorphisms of H^∞ , and zero multiplicities. Exact-title, exact-conjecture, arXiv-id-plus-correction, and power-pullback searches were run in the local parsed arXiv corpus and arXiv-indexed web results through 2026-08-11. No later correction or prior answer was found. Novelty confidence is medium because the argument is elementary. Human review should focus on the source’s explicit allowance of nonclosed subalgebras and on the conjugacy formula defining $T_{a,m}$.

References

- [1] K. Behera, R. Maurya, and P. Muthukumar, *Automorphisms of subalgebras of bounded analytic functions*, arXiv:2412.03245 (2024; revised 2025), concluding conjecture on p. 16.